

124104069

NIT KURUKSHETRA: END-SEMESTER

Course No.: CSIC-221 Course Title: ML and Data Analytics
Duration: 180 Mins Date: 27-11-2025

Max. Marks: 80
Time: 09:30 AM - 12:30 AM

Note 1: Read and FOLLOW Instructions to candidates on 1st page of your answer script.
Note 2: Answer all subparts of a question together.

Q1: Assume the following dataset is given: (1,2), (1,4), (5,1), (6,3), (8,5), (9,9), (2,4), (4,2).
K-Means is used with $k=3$ to cluster the dataset. Moreover, Euclidean distance is used as the distance function to compute distances between centroids and objects in the dataset. Moreover, K-Means's initial clusters C1, C2, and C3 are as follows: [6+6= 12Marks]

C1: {(1,2), (1,4), (6,3)}

C2: {(2,4), (4,2)}

C3: {(5,1), (9,9), (8,5)}

- Now K-means is run for a two iteration; what are the new clusters and what are their centroids?
- Perform single-link hierarchical clustering method and show the results using a dendrogram.

Q2: Consider the following dataset [2+2+2=6 Marks]

- A = {80, 16, 96, 10, 10, 35, 10, 10, 30, 10, 10, 39, 10, 16, 33 }

- Calculate 25%, 50%, 75% for the above Dataset.
- Calculate the standard deviation and variance for the above Dataset.
- Use the data to create a box-and-whisker plot.

- or

- Above Standardization on above dataset

Q3 a: Prove that A (1,2,6,5,7) and B(4,6,9,6,8) are correlate or not using Pearson correlation coefficient value. [4 Marks]

Q3 b: Compare between batch gradient descent, Mini-batch gradient descent and stochastic gradient descent methods. [4 Marks]

Q3 c: Write down the procedure to compare two different clustering techniques. [4 Marks]

Q4 a: With the following data set predict class label of point (60,40) and for point (45,45) with KNN method with $k=3$. [6 Marks]

A1	40	50	60	10	70	60	25
A2	20	50	90	25	70	10	80
Label	C1	C2	C2	C1	C2	C1	C2

Q4 b: Prove that the mean of $n1 = \frac{(a+b)}{2}$ and $n2 = \frac{(a-b)}{2}$ is $\frac{(a)}{2}$, and the variance is $\frac{(b * b)}{4}$ [4 Marks]

Q4 c: The logistic function is given by $\sigma(x) = \frac{1}{1+e^{-x}}$ Show that $\frac{\partial(\sigma(x))}{\partial x} = \sigma(x) * (1 - \sigma(x))$. [4 Marks]

Q5: Consider the data set shown in below Table:

Sl No.	Expr	NOI	KNO	NOP	Inc
A	3	2	1	1	1
B	5	5	1	2	1
C	1	3	4	1	1
D	2	3	5	4	1
E	4	5	2	5	0
F	1	1	3	2	1
G	2	2	3	2	1
H	1	5	2	2	0
I	4	4	3	4	0
J	5	1	1	5	0

Q5 a. Compute 95% confidence interval of Features: Expr and NOP. [6 Marks]

Q5 b. Compute the value of gini split or information gain for all features and find best sets of features using feature ranking concept. Note: use mean to find high and low value [6 Marks]

Q5 c. Compute conditional probabilities for Consider the data set shown in above Table. find out class for the following test data using naïve Bayes classifier. [6 Marks]

* Expr=1, NOI=3, KNO=5, NOP=3

* Expr=2, NOI=1, KNO=3, NOP=4

Q6: A placement division NIT KKR opinion poll surveyed a simple random sample of 1000 students. Respondents were classified by department (CSE or EC) and by voting preference (Microsoft, Apple, Oracle). Results are shown in the contingency table below.

	Microsoft	Apple	Oracle	Row Total
CSE	200	150	50	400
EC	250	300	50	600
Column total	450	450	100	1000

Do the CSE voting preferences differ significantly from the EC preferences? Use Chi-Square Test with 0.05 level of significance. [6 Marks]

Q7: Table shows the actual and predicted value of the developed model.

i Construct the confusion matrix for Table 2 [2 Marks]

ii Compute Accuracy, Recall, Precision and F-Measure. [4 Marks]

Actual	1	1	0	0	0	1	1	0	0	0	1	0	1	1	1
Predicted	1	0	1	1	1	1	1	1	1	1	1	0	0	0	1

Q8: Apply linear regression with Stochastic Gradient Descent on below data for one iterations. Assume m and c as 0 in initial step and learning rate as 0.1. [6 Marks]

X	1	1	2	3	4	5	2	3
Y	1	2	3	2	3	6	3	4